Stage	Program	Block	Description	Qualitative	Quantitative	Output
Raw data evaluation			- getting to know your data, GTKYD	Yes	Yes	
	AFNI GUI		- visualize data images, time series graphs, and more	- align: Check alignment (or registration) features. - graph: View the time series plots of one or more voxels. - instacorr: Flag peculiar spatio-temporal patterns in the time series data. - other: Any other feature(s) using the afni and/or suma GUIs, plugins, etc.		
	gtykd_check.py		- summarize header and data (min/max) properties - verify desired properties for one or more datasets - evaluate consistency of properties across group		- header-info: Matrix size, orientation, voxel dimensions, datum type, NIFTI qform_code, NIFTI sform_code - data-info: Number of runs, minimum value, maximum value. - EPI header-info: TR, number of time points, slice timing.	
Prepressing – Structure T1*	sswaper2		- bias field correction & grad unwarp	- Field of view (FOV). - Distortion. - High resolution and contrast. - Alignment is good: tissue boundaries and sulcal and gyral patterns appear to be well-aligned.		In the T1 results folder, AM*.jpg, MA*.jpg, QC_anatQQ*.jpg and QC_anatSS*.jpg are the images that we care most. In the ssw_aligh_hist.* folder you will see the step-by-step images generated in the process, not very necessary.
Preprocessing - APQC HTML	**afni_proc.py or ap_run_simple_rest.tcsh:** gen_epi_review.py (@epi_review), gen_ss_review_scripts.py (@ss_review_basic, @ss_review_driver, and @ss_review_driver_commands), apqc_make_tcsh.py, apqc_make_html.py, run_qc_*.tcsh		- evaluate systematic images and quantities (alignment, regression, etc.) - run interactive data exploration in GUIs - save QC ratings and commentsGroup summary tables (of afni_proc.py and more)	Yes	Yes	QC_${subj} directory: out.ss_review.$subj.txt, out.ss_review_uvars.json 

```
subject ID                 : 091
AFNI version               : AFNI_99.99.99
AFNI package               : Linux_cmake
TR                         : 1.0
TRs removed (per run)      : 0
multiband level            : 1
slice timing pattern       : simult
num stim classes provided  : 3
final anatomy dset         : anat_final.091+tlrc.HEAD
final stats dset           : stats.091_REML+tlrc.HEAD
orig voxel counts          : 102 102     64
orig voxel resolution      : 2.509800    2.509800     2.499786
orig volume center         : -18.062111  -15.281898   -11.606758
final voxel resolution     : 2.500000    2.500000     2.500000

motion limit               : 0.3
num TRs above mot limit    : 5
average motion (per TR)    : 0.103108
average censored motion    : 0.0993947
max motion displacement    : 1.64944
max censored displacement  : 1.63683
outlier limit              : 0.05
```


```
{
  "afni_package": "Linux_cmake",
  "afni_ver": "AFNI_99.99.99",
  "align_anat": "091_al_keep+orig.HEAD",
  "censor_dset": "censor_091_combined_2.1D",
  "copy_anat": "anatSS.s091+orig.HEAD",
  "cormat_warn_dset": "out.cormat_warn.txt",
  "df_info_dset": "out.df_info.txt",
  "enorm_dset": "motion_091_enorm.1D",
  "errts_dset": "errts.091_REML+tlrc.HEAD",
  "final_anat": "anat_final.091+tlrc.HEAD",
  "final_epi_dset": "final_epi_vr_base_min_outlier+tlrc.HEAD",
  "final_view": "tlrc",
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  "gcor_dset": "out.gcor.1D",
  "have_radcor_dirs": "yes",
  "mask_anat_templ_corr_dset": "out.mask_at_dice.txt",
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  "mask_dset": "mask_epi_anat.091+tlrc.HEAD",
  "max_4095_warn_dset": "out.4095_warn.txt",
  "mb_level": 1,
  "mot_limit": 0.3,
```
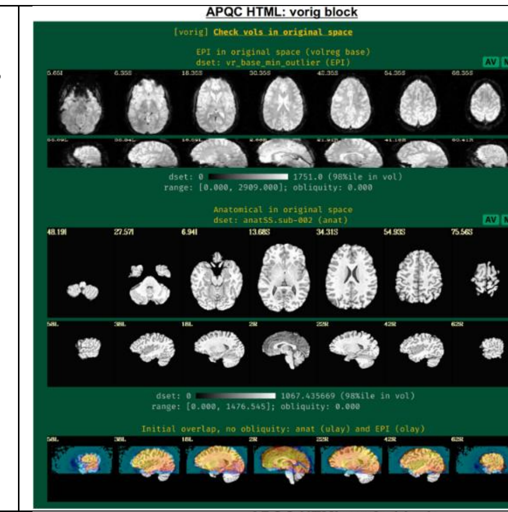
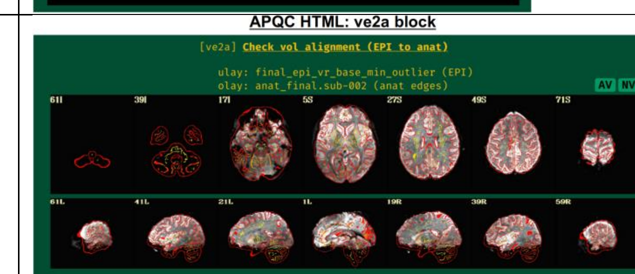
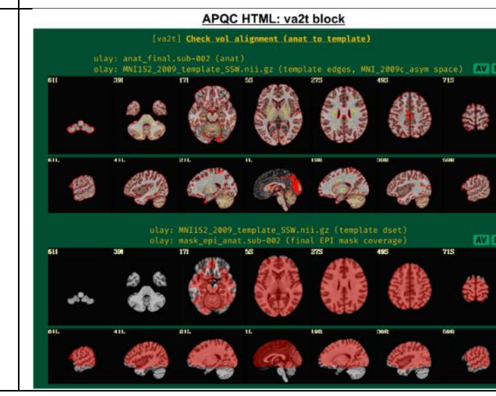
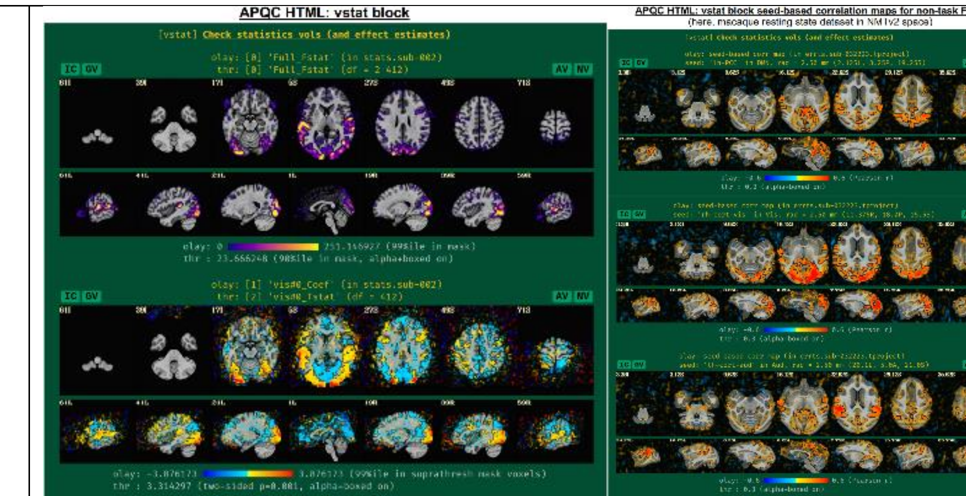
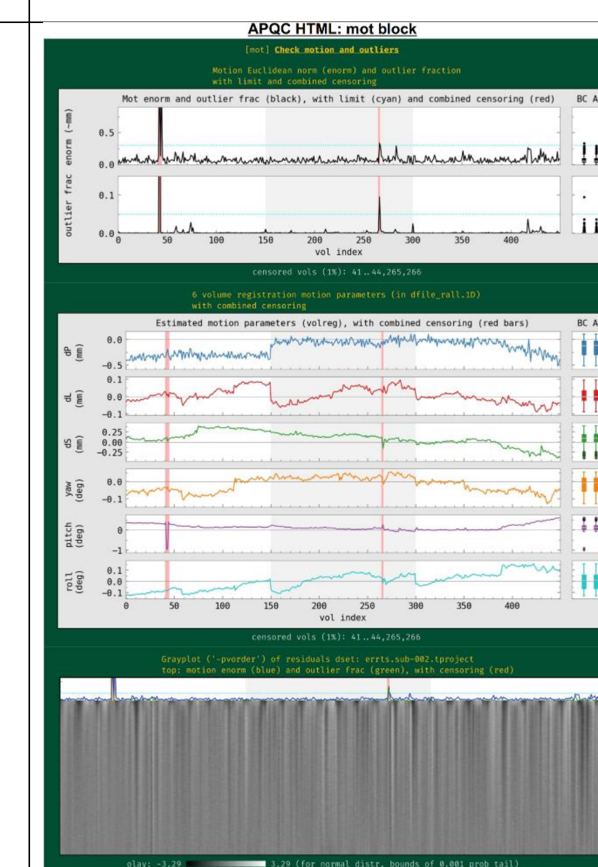
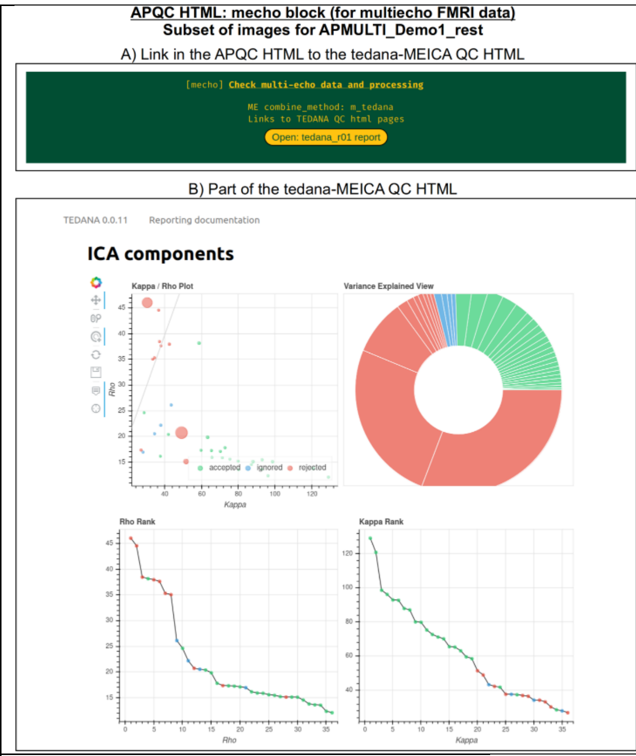
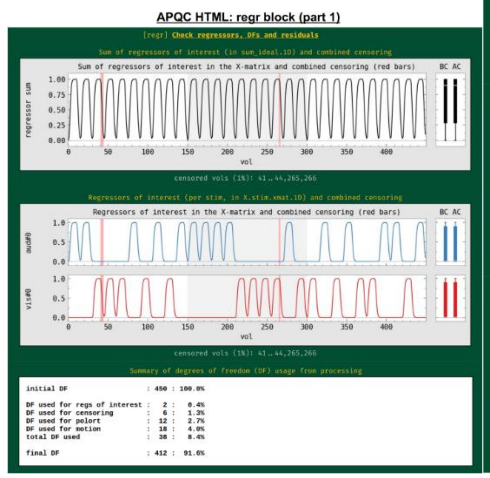
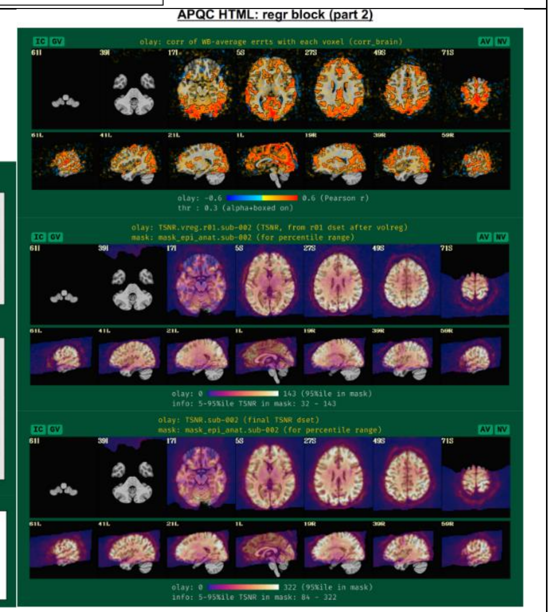


Qiuyu yu. https://github.com/QiuyuYu3

	gen_ss_review_scripts.py	**Vorig**	- EPI and anatomical FOV coverage, distortion, tissue contrast, brightness inhomogeneity. - Initial EPI-anatomical overlap	- The entire brain is in the FOV (no missing region) - No severe brightness inhomogeneity or distortion - appropriate tissue contrast - No flip - Initial EPI-anatomical overlap (alignment, bottom row) is not far away from each other	- The grayscale colorbar does not contain negative values (indicating potential issues with data acquisition or conversion) and that the maximum value does not exceed 4095 (suggesting saturation of the 12-bit intensity signal).	
		ve2a	- **EPI-anatomical alignment** - EPI coverage, distortion	- Good alignment: similar internal structures in the same location, such as ventricle edges, tissue boundaries, and gyral/sulcal patterns. In judging EPI-anatomical alignment, **interior structures matter** most and CSF should be ignored. - Poor: low or nonstandard tissue contrast. - Good signal coverage (any missing region or distortion in functional ROI?)		
		va2t	- **anatomical-template alignment** - Final EPI coverage in template: the EPI mask[1] (calculated as the intersection of the anatomical and EPI data) is aligned with the template.	- Good alignment (e.g., from sulcal, gyral and ventricle patterns). - EPI coverage is relatively strong or potentially missing.		

[1] Mask-overlap: Estimated coverage of usable FMRI signal (typically intersected with the subject anatomical).

		Vstat	- effect estimate + statistics maps, correlation maps, artifacts - expected/unexpected patterns (via InstaCorr, Graph View, etc.)	- Any scanner artifacts in the EPI data (ghosting; striping; strong slice-based patterns; large spatial patterns across/unconstrained by tissue type; notably non-physiological patterns.), stimulus-correlated subject motion, poor task compliance by the subject, incorrect timing in the stimulus files, or other sources. - Colorbar: values above threshold are opaque and outlined. Some extraneous noise features may be outside the brain because of ghosting, bad coil features, poor alignment, or some other pathology in the data. - Enough F-stat values (good fits) localized in grey matter. - Not too scattered, not noisy, or low in some/many expected ROIs		 *For non-task datasets, such as resting state, a set of seedbased correlation maps is displayed if the template is in a recognized space (you can choose your own seed).*
		Mot	- subject motion, stimulus-correlated motion - "**enorm**" parameter (the Euclidean norm of the forward differences of the **six rigid body motion parameters**, estimated via volume registration) and the "**outlier fraction**" (the fractional count of voxels in a brain mask that are **outliers** in their time series).	- Note red vertical bands(outlines), boxplot [before censoring (BC) and after censoring (AC)] - a "gray plot" or "carpet plot" (3dGrayplot) of FMRI time series across the brain can be useful for visualizing nonlocal/global features in time series: each time series is represented as a fluctuation in grayscale coloration and displayed by z-value. Strong vertical patterns may suggest high residual correlation (primarily checked in "regr-corr_brain" visualization and quantified in "qsumm" with GCOR).	- the user-specified censor limits - duration of each run - censored volumes	

		Mecho	- multiecho FMRI features and processing, to **boost local SNR and potentially reduce artifacts** by using **optimal combination** (OC) calculation and the original **MEICA** algorithm.	Yes	Yes	
		Regr	- spatiotemporal structure in the residuals, corr_brain, (or “error time series”, errts), total degrees of freedom, and TSNR.	- idealized stimulus responses and the censored motion outliners. - task-ideal-sum: (NB: Strongly paradigm dependent) any problem with the sum of regressors; large gaps and/or spikes might generally be worth noting. - task-ideal-stim: (NB: Strongly paradigm dependent) any problem with an individual regressor of interest; duplicated stim timing (scripting mistake); stimulus-correlated motion may be worth noting. - **corr_brain map** (spatial correlation structure remaining in the residuals) is the correlation of each voxel’s residual time series with the average time series of all residuals across the brain (which might be called the “global residual”): - large negative correlations or high magnitude patterns (high correlations) that extend across	- Do not use too many degrees of freedom, model will be saturated. Motion regressors, slow-drift regressors, censoring, and bandpassing are common model components that reduce DFs - **TSNR plots:** TSNR we care about can be different at different stages. *For example, TSNR is first calculated directly after volume registration (middle plot), and the “signal” is the (full) time series and the “noise” is the detrended time series; the second TSNR calculation comes after regression modeling (bottom plot), and the “signal” is the full (all_runs) time series and the “noise” is the errts time series.* - TSNR_volreg: Mainly informational, since this is calculated before regression modeling and noise regression (look for similar features as in TSNR_final-* items). - TSNR_final-loss: Notable dropout/low signal in regions of interest (e.g., often low in frontal/temporal lobes and subcortical nuclei).	 

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non-GM (gray matter) tissues can be scanner issues, motion-related artifacts, or having too few DFs left in the time series.
- ROIs in template: enough voxels.
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- TSNR_final-artifact: Non-physiological patterns of TSNR magnitude, particularly dropout (e.g., vertical bands/stripes).

- ROI shape and TSNR stats table: make sure the ROI has enough signal and voxels if the whole TSNR is low.
- Nvox, total number of voxels in ROI (warnings if quite small)
- Nzer, number of zerovalued voxels (warnings as the fraction Nzer/Nvox increases)
- Dvox, the maximum depth of this ROI, counted in voxels (warnings as this decreases)
- Tmin, T25%, Tmed, T75%, Tmax, which are the min, max, and quartile values of TSNR (warnings as T75% decreases, and as the slope of values (T75%—T25%)/Tmed increases)
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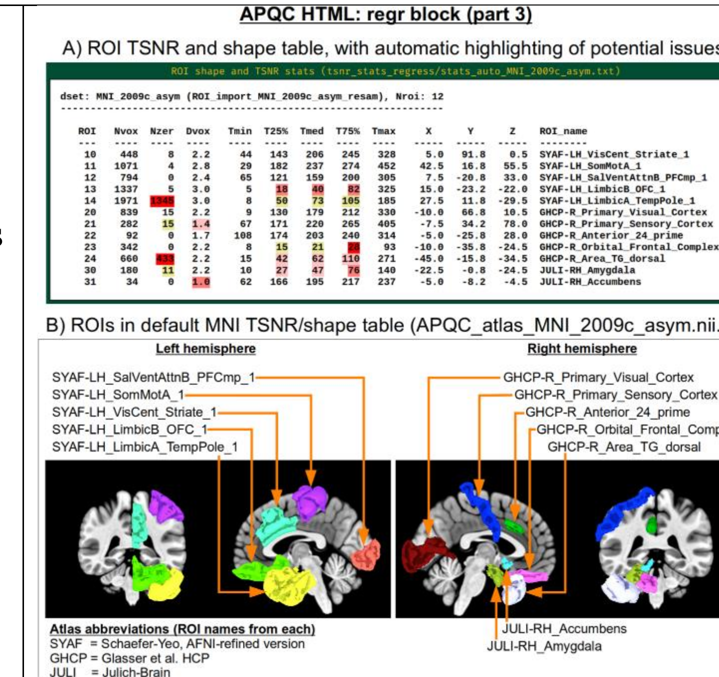

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<td>@radial_correlate</td>
<td>radcor</td>
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- local correlation patterns across the brain, InstaCorr of EPI at different stages
- @radial_correlate calculates radial correlation (radcor) of each voxel in a dataset; that is, the correlation with a Gaussian-weighted average of the voxels in a large radius (full width at half maximum (FWHM) of 20 mm, by default) around it to detect certain kinds of coil artifacts or motion, which can produce large, highly correlated patches across the brain.
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- As long as you indicate in afni_proc.py, AFNI will make radcor at each stage (e.g. after processing a head movement). Make sure to check the radcor values both before and after volume registration to reduce the effects of subject motion.
- Motion: high correlation following the edge of the brain or EPI FOV.
- Bad coil: anterior and right octant of the brain is extremely highly correlated.
- In regression stage, seeing high radcor values constrained within GM can be a positive indication that both motion effects are reduced (because those would likely produce correlations across tissue boundaries) and that task response is high (because that would likely increase correlations across functionally related GM).
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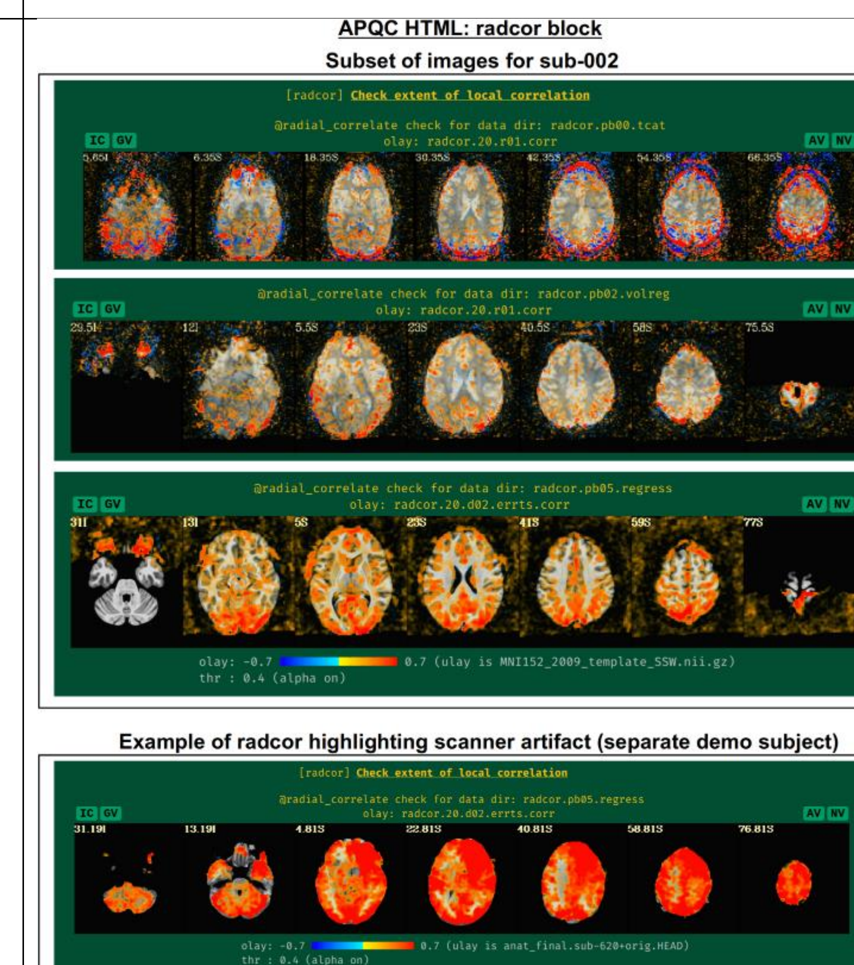

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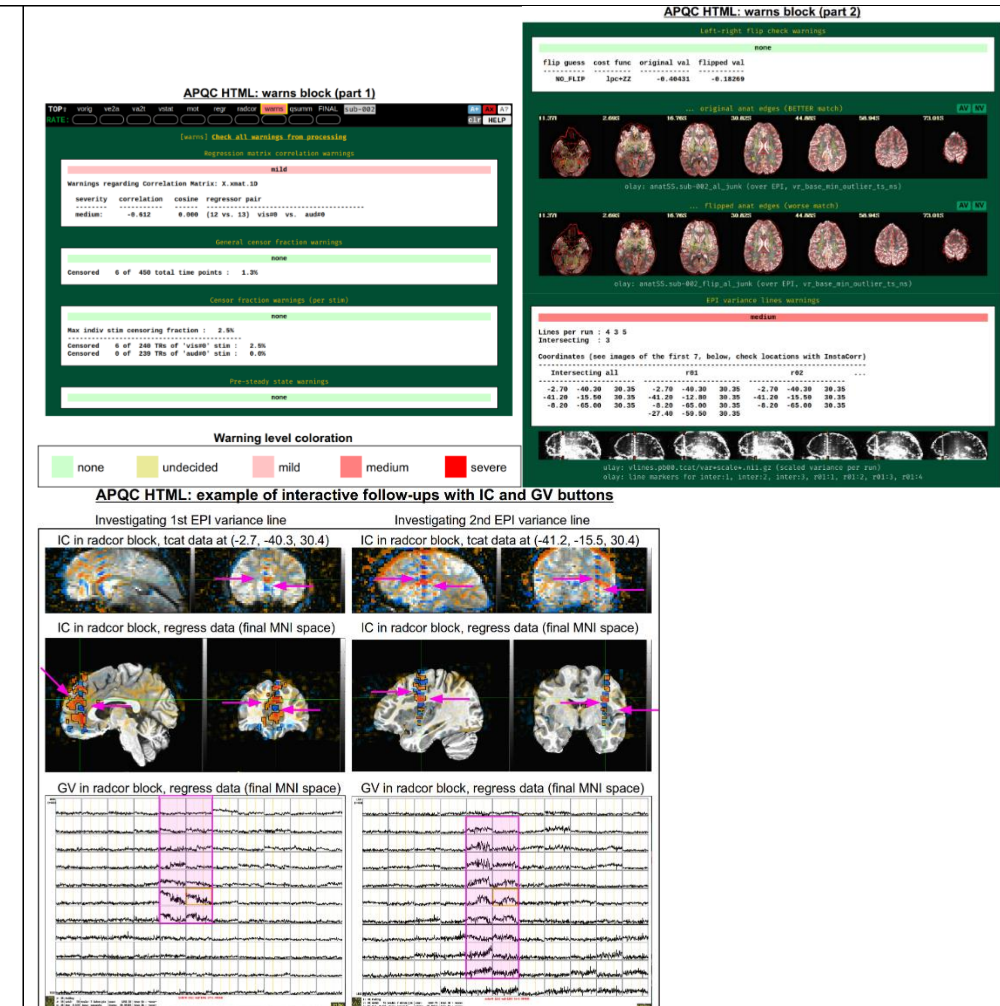
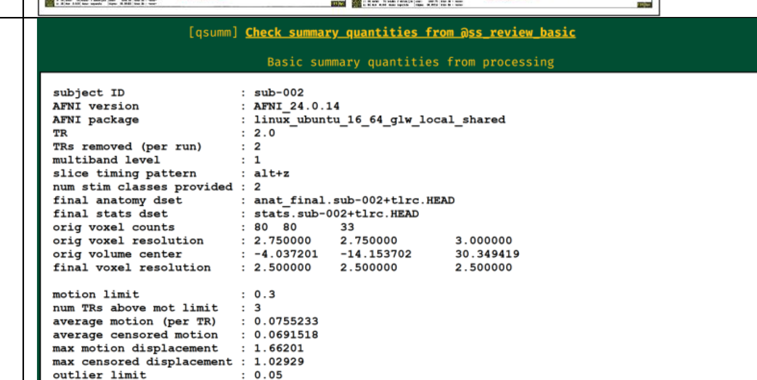
		warns	presteady state, left-right flip, variance artifacts, high subject motion, regression correlation, etc. https://www.biorxiv.org/content/10.1101/308643v1.full.pdf	- check for **left-right flip mismatches** between the **EPI and anatomical datasets** ("-align_opts_aea -check_ flip") - Artifacts: **narrow lines of high variance** that are orthogonal to slice planes in EPI. You can then click on IC in the radcor block's "regress" image and navigate to the equivalent anatomical location to view the artifact's effect after regression modeling.	- regr_mat: High pairwise correlations in regression matrix (varied). - gen-censor: High total/overall censoring fraction (typically checked with GSSRT). - task-stim-censor: High censoring fraction for one or more particular stimuli (optionally checked with GSSRT). - press: EPI data appear to have pre-steady state volumes at the start (via outlier check; though sometimes this is simply due to motion in first time points).	
	@ss_review_basic	**qsumm**	[quantitative summary values](#). *These specific threshold values that used for the quantitative keys could differ across studies (thinking about ADHD children).*		- Final DF fraction >0.7 - Censor fraction < 0.15 - Average censored motion <0.1 - Max censored displacement <6 - Global correlation (GCOR) <0.15 - Flip guess - TSNR >= 150 (warning)	[qsumm] Check summary quantities from @ss_review_basic Basic summary quantities from processing subject ID : sub-002 AFNI version : AFNI_24.0.14 AFNI package : linux_ubuntu_16_64_glw_local_shared TR : 2.0 TRs removed (per run) : 2 multiband level : 1 slice timing pattern : alt+z num stim classes provided : 2 final anatomy dset : anat_final.sub-002+tlrc.HEAD final stats dset : stats.sub-002+tlrc.HEAD orig voxel counts : 80 80 33 orig voxel resolution : 2.750000 2.750000 3.000000 orig volume center : -4.037201 -14.153702 30.349419 final voxel resolution : 2.500000 2.500000 2.500000 motion limit : 0.3 num TRs above mot limit : 3 average motion (per TR) : 0.0755233 average censored motion : 0.0691518 max motion displacement : 1.66201 max censored displacement : 1.02929 outlier limit : 0.05 average outlier frac (TR) : 0.00276734
Final assessment and visualization	gen_ss_review_table.py + \`extra_info/out.ss_review.${subj}.json\` or \` apqc_${subj}.json\`		- summarize a group review of both raw and processed data properties by using \`out.ss_review.${subj}.txt\` input files. - list subjects with specific data properties (e.g., drop criteria).		The rating results from APQC HTML will be added in \`extra_info/out.ss_review.${subj}.json\` or \` apqc_${subj}.json\`, you can combine the results from the json file to \` gen_ss_review_table.py\`	

Reference:

https://afni.github.io/qc-demo-repo/
Reynolds, R. C., Taylor, P. A., & Glen, D. R. (2023). Quality control practices in FMRI analysis: Philosophy, methods and examples using AFNI. Frontiers in Neuroscience, 16, 1073800. https://doi.org/10.3389/fnins.2022.1073800
Paul A. Taylor, Daniel R. Glen, Gang Chen, Robert W. Cox, Taylor Hanayik, Chris Rorden, Dylan M. Nielson, Justin K. Rajendra, Richard C. Reynolds; A Set of FMRI Quality Control Tools in AFNI: Systematic, in-depth, and interactive QC with afni_proc.py and more. Imaging Neuroscience 2024; 2 1–39. doi: https://doi.org/10.1162/imag_a_00246